

Aditya Acharya

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SUMMARY

Software engineer specializing in containerized backend systems and AI-enabled applications. Built Flask-based simulation APIs for a graph-driven supply-chain platform, supporting 1,000+ timesteps per scenario, and developed React + Three.js analytics dashboards that reduced manual analysis time by 40%. Designed end-to-end agentic AI chatbot architectures for financial services, with robust orchestration and error handling across complex edge cases. Seeking to apply backend, AI, and full-stack expertise to build scalable, data-driven systems with measurable product impact.

EDUCATION

University of Illinois, Chicago
Master of Science in Computer Science

May 2024 - May 2026
CGPA: 3.8

SVKM Narsee Monjee Institute of Management and Studies
Bachelor of Engineering, Computer Science with Specialization in Artificial Intelligence

May 2020 - May 2024
CGPA : 3.67

TECHNICAL SKILL

- **Skills:** Python, JavaScript, TypeScript, EDA, Tableau, Pandas, NumPy, JavaScript, C/C++, React JS, Node JS, MongoDB, SQL, MySQL, PostgreSQL, Postman API, AWS, GCP, Firebase, Agile, Data Analytics, VMWare, PowerBI, Snowflake, Risk Management
- **Technology Tools:** Git, GitHub, Android Studio, VS Code, Linux, Wireshark, Figma, Slack, MS Excel, MS Access, Illustrator

EXPERIENCE

Delta Labs AI
Software Engineering Intern

May 2025 - Dec 2025
Nashville, TN

- Containerized backend services using Docker, cutting environment setup time from hours to <15 minutes and ensuring reproducible deployments across development machines
- Designed and deployed Flask-based simulation APIs for a graph-driven supply chain system, supporting 100+ nodes and edges per run with deterministic stepwise execution for experimentation and analysis.
- Implemented a state introspection and step execution pipeline to simulate 1,000+ discrete timesteps per scenario, significantly improving debugging efficiency and system validation.
- Built React + D3.js visual analytics dashboards with 10+ interactive views, reducing manual analysis time by 40% for internal stakeholders.
- Prototyped an agentic AI chatbot (OpenAI API) mapping natural-language supply chain queries to backend simulations.

Kotak Securities
AI System Design & Product Intern

Dec 2023 - Jun 2024
Mumbai, IND

- Evaluated 8+ competitor stock trading chatbots, testing 50+ NLP query permutations to assess intent handling, ambiguity resolution, and failure behavior.
- Designed an end-to-end agentic AI chatbot architecture, defining 5+ system modules to improve robustness, reasoning flow, and response reliability.
- Specified error-handling logic for 10+ edge cases, improving transparency and user trust in AI-generated financial insights.
- Documented system workflows and decision logic for the agentic chatbot, enabling smoother handoff to engineer for implementation.
- Translated compliance and ambiguity risks into AI guardrails, reducing high-risk financial responses by ~30%

PROJECTS

Visualizing Speech and Biomarker Correlations in Alzheimer's Disease
UIC AI.Health4All

Aug 2025 - Dec 2025

- Built an interactive visualization system for the SLaCAD dataset (91 participants, 7.5+ hours of speech, 329 multimodal features) integrating linguistic, acoustic, biomarker, and clinical data.
- Applied ML-based feature selection (Random Forest, ANOVA, Mutual Information) to identify the top 10 predictive features, achieving 94.7% accuracy for Normal vs Impaired (AD/MCI) classification.
- Developed React + D3.js visualizations (dynamic Pearson correlation matrices, TF-IDF word clouds) revealing 12.8× higher lexical diversity in Normal participants and enabling clinician-driven exploration

AI based Hippocampal Contouring for Whole Brain Radiotherapy
Nanavati Hospital Max

Oct 2021 - Feb 2022

- Developed a CNN-based medical imaging pipeline for hippocampus brain tumor detection using 1,000+ DICOM MRI scans.
- Migrated preprocessing and training workflows from MATLAB to Python, improving maintainability and integration with modern ML tooling.
- Optimized CNN architectures and training loops through 1,500+ experimental iterations, reducing training time by ~20% while improving classification performance.